

HW #18

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Prob: 1:

$$a) \mathbb{T} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ x-y \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2x \\ x-y \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{array}{l} 2x=0 \Rightarrow x=0 \\ x-y=0 \Rightarrow y=0 \\ 0=0 \Rightarrow 0=0 \end{array}$$

unique

So it is one-to-one

$$\mathbb{T} \begin{bmatrix} 2x \\ x-y \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \text{Not possible since it can only map in } \mathbb{R}^3$$

Not onto.

$$b) \overbrace{D(p(x)) = p'(x)}$$

$$p'(x) = 0 \quad \text{iff a constant}$$

since $P_3 \rightarrow P_2$

It is not one-to-one.

~~$p'(x)$~~ $P_2 \rightarrow P_3$ by integration we can have polynomial of degree 3. Range is entire P_3 .
It is onto.

c) $A^T = 0 \iff A = 0$
 $\ker(A) = 0$
 it is one-to-one.

for all A matrices

$$T(A) = A^T$$

$$T(A^T) = A$$

therefore it is onto.

Problem 2:

$$T \begin{bmatrix} a \\ b \end{bmatrix} = a + (a+b)x$$

$$a + (a+b)x = 0$$

$$a = 0$$

$$\begin{matrix} a+b=0 \\ b=0 \end{matrix} \Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\ker(T) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

it is one-to-one

$$T(v) = p(x)$$

$$a + (a+b)x = c + dx$$

$$a = c$$

$$b = d - c$$

$$\Rightarrow \begin{bmatrix} c \\ d-c \end{bmatrix}$$

Problem 3:

Since the transformation is both one-to-one & onto, By theorem it is invertible.

Problem 4:

$$T(x_1, x_2, x_3) = \begin{bmatrix} 3 & -1 & 0 \\ -2 & -4 & 4 \\ 5 & 1 & -2 \end{bmatrix}$$

$$\begin{aligned} |T| &= 3(-4) - 1(-11) \\ &= -12 + 11 \\ &= -1 \end{aligned}$$

$$\text{Adj } A = \begin{bmatrix} -4 & 11 & 18 \\ 2 & -6 & -7 \\ 3 & -9 & -10 \end{bmatrix}^t = \begin{bmatrix} -4 & 2 & 3 \\ 11 & -6 & -9 \\ 18 & -7 & -10 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} 4 & -2 & -3 \\ -11 & 6 & 9 \\ -18 & 7 & 10 \end{bmatrix}$$